



Nicolas Anderson

Graduate Student – Geology and Planetary Science
California Institute of Technology
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Education

California Institute of Technology – Pasadena, CA	2023 – present
Graduate Student – Geology and Planetary Science Using field observation, paleomagnetism, geochemistry, geochronology, and petrology to better develop our understanding of igneous intrusions, magma evolution and Earth's magnetic field.	
University of California, Berkeley – Berkeley, CA	2020 – 2022
B.A. in Geology with Honors Honors Thesis: <i>Paleomagnetism & Geochronology of Mesoproterozoic Diabase Sills from Death Valley, California: Evaluating the Bifurcating Plume Hypothesis for the Southwestern Laurentia large igneous province</i> ; Advised by: Nicholas Swanson-Hysell, PhD & Eben Blake Hodgins, PhD <i>Relevant Coursework:</i> Mineralogy, Petrology, Structural Geology, Aquatic Geochemistry, Isotopic Geochemistry, Soil Science, Geomorphology, Field Geology & Mapping, Stratigraphy, Paleomagnetism, Carbon Cycle Dynamics	
City College of San Francisco – San Francisco, CA	2016 – 2020

Teaching Experience

Co-instructor for Advanced Field Course – Nanyang Technical University, Singapore	2023
Taught and guided 20 geology students from Nanyang Technical University, Singapore on a 3-week camping field geology campaign in eastern California focused on: complex fold and thrust structural mapping in Cerro Gordo; glacial geomorphology mapping and fault scarp offset surveying in Lundy Canyon; Sighting, surveying, and topographic map generation in the Alabama Hills.	

Work Experience

Research Assistant – Earth Surface Dynamics Laboratory (Michael Lamb), Caltech – Pasadena, CA	2023
Design, fabricate and test clay and sediment flocculating devices for use in projects conducted by SURF and WAVE undergraduate fellows. Rewire and setup defunct lab equipment.	
Exploration Geologist – Brimstone Energy – Oakland, CA	2022 – 2023
Use geoscientific datasets to identify and prioritize exploration targets. Mapping, sample collection and report synthesis for mafic intrusions and metamorphic belts across the US for production of carbon neutral Portland cement substrates.	
Undergraduate Research Assistant – Swanson-Hysell Lab Group, UC Berkeley – Berkeley, CA	2020 – 2022
Cutting, drilling, orienting, and measuring paleomagnetic samples on SQUID DC2 for plate reconstructions. Magnetic and heavy liquid mineral separation, picking and mounting of zircons. SEM and XRD to determine mineral phases.	

Presentations

GSA 2022 – Joint Section Cordilleran/Rocky Mountain Meeting – Las Vegas, NV	2022
Oral Presentation: <i>Paleomagnetism & Geochronology of Mesoproterozoic Diabase Sills from Death Valley, California: Evaluating the Bifurcating Plume Hypothesis for the Southwestern Laurentia large igneous province</i>	
EPS Undergraduate Honors Symposium – Berkeley, CA	2022
Oral Presentation: <i>Paleomagnetism & Geochronology of Mesoproterozoic Diabase Sills from Death Valley, California: Evaluating the Bifurcating Plume Hypothesis for the Southwestern Laurentia large igneous province</i>	
Summer Undergraduate Research Fellowship Conference – Berkeley, CA	2021
Oral Presentation: <i>Testing the hypothesis of a bifurcating mantle plume 1.1 billion years ago with geochronology and paleomagnetism of the Southwestern Laurentia large igneous province.</i>	

Publications

- Zhang, Y., **Anderson, N.S.**, Mohr, M.T., Schmitz, M.D., Macdonald, F.A., Nelson, L.L., Thurston, O.G., Guenthner, W.R., Karlstrom, K.E., Swanson-Hysell, N.L. *Paleomagnetic Records From Pulsed Magmatism in the Southwestern Laurentia Large Igneous Province and Cardenas Basalt Support Rapid Late Mesoproterozoic Plate Motion*. *JGR:Solid Earth*, 2024. doi: 10.1029/2024JB029036. 2024
- M.T. Mohr, M.D. Schmitz, N.L. Swanson-Hysell, K.E. Karlstrom; F.A. Macdonald, M. Holland, Y. Zhang, **N.S. Anderson** - *High-Precision U-Pb geochronology links magmatism in the SW Laurentia Large Igneous Province and Midcontinent Rift* 2024

Field Work

- Colbalt, ON, Canada** 2023
Used paleomagnetic studies to develop a model of the timing of folding and carbonate vein intrusion associated with 5-element vein deposits for economic geology.
- Columbia River Gorge, OR** 2022
Investigated post-rainstorm, fire-prone, high-slope regions to better quantify debris flow quantity and frequency along high-risk Columbia River Gorge catchments. Scouted for potential landing zones for 1m resolution drone LIDAR.
- Point Lobos, CA** 2022
Determined the relationship, tectonic history, and post-deposition deformation between outcropping Salinian igneous rock and the overlying sediment using field relationships and sedimentary lithofacies stratigraphic sections.
- Benton, CA** 2022
Large scale mapping of the Benton Range to constrain the tempo of dike emplacement and subsequent tectonic folding as it pertains to crustal stresses and magmatism associated with the Nevadan orogeny along Laurentia's western coastline.
- Mono, CA** 2022
Surficial glacial geomorphology mapping to characterize landscape evolution and extent of glaciation since the LGM. Slope surveys of fault scarps of well-dated Sierran moraines to determine average rate and direction of slip for Hilton Creek Fault since the LGM.
- Death Valley, CA** 2021
Field mapping, section measurement, and sample collection to develop stratigraphic, geochronologic and paleomagnetic data for Southwestern Laurentia large igneous province.

Funding/Awards

- Ian Campbell Award (for outstanding performance in field geology courses)** 2025
- National Science Foundation Graduate Research Fellowship Program (NSF GRFP)** 2023
- Shoemaker Impact Cratering Award** 2023
- Caltech Internal Seed Funding** 2023
- Caltech Provost Diversity Fellowship** 2023
- Summer Undergraduate Research Fellowship** 2021
- Charles H. Ramsden Endowed Scholarship** 2021

Skills

Computing: ArcGIS, QGIS, Photoshop, Illustrator, GitHub, Pmag GUI, Microsoft Visual Basics, Python, HTML, LaTeX
Geology: SQUID DC2 & OPRM Magnetometers, Frantz magnetic mineral separation, heavy liquid mineral separation, XRD, SEM
First Aid: NOLA Wilderness First Aid Certified

Service and Outreach

- San Francisco Bike Coalition** – San Francisco, CA 2016 – 2022
Encourage Bay Area residents to advocate for bicycle legislature. Provide bike parking valet services at large events & offer bike mechanic services for community bike builds.
- Bay Area Ridge Trail / Billion Oyster Project / Million Trees NYC** 2008 – 2022
Trail maintenance, oyster reintroduction, invasive species removal, watershed, and habitat restoration
- Earth Science Tutor** – City College of San Francisco 2016 – 2020
Subjects: Oceanography, Geography, Geology